

Foundations of the Canonical MASA–Stack Class

A rigorous proof appendix for the geometric/cohomological layer of Paper A

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Abstract

This appendix supplies rigorous foundations for the geometric and cohomological constructions of Paper A. We (A) fix the objects — the space X of maximal abelian subalgebras (MASAs) of $M_n(\mathbb{C})$ as the homogeneous space $\mathrm{PU}(n)/\mathrm{PN}(T)$, the action groupoid $\mathrm{PU}(n) \ltimes X$ presenting the stack $[X/\mathrm{PU}(n)]$, and a single, explicitly named cohomology theory that is never silently switched; (B) construct the canonical class from $1 \rightarrow \mathrm{U}(1) \rightarrow \mathrm{U}(n) \rightarrow \mathrm{PU}(n) \rightarrow 1$ and prove its vacuity on $\mathrm{U}(n) \ltimes X$ and its nontriviality on $\mathrm{PU}(n) \ltimes X$; (C) prove the order is exactly n , giving both $n[\alpha] = 0$ (determinant-twist splitting) and $k[\alpha] \neq 0$ for $1 \leq k < n$, with the role of the S_n -equivariant homomorphism lemma made explicit and the “exotic splitting” objections addressed; (D) state and prove the Morita-equivalence step, naming the theorem and verifying its hypotheses; and (E) compute $H^2(\mathrm{PN}(T); \mathrm{U}(1)) \cong \mathbb{Z}/n \oplus \mathbb{Z}_2$ ($n \geq 4$) via a Lyndon–Hochschild–Serre spectral sequence with all differentials, extension problems, coefficient conventions, topology assumptions, and low- n exceptions recorded. **Intellectual honesty is the point:** every result is tagged COMPLETE, SKETCH (reduced to a cited theorem whose hypotheses we verify), or GAP (an explicit open sub-lemma), and a status table closes the document.

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Reading guide and honesty protocol

Each numbered result carries a **Status** line immediately after its proof:

- **COMPLETE PROOF:** a full argument, checkable by a specialist in equivariant topology / stack cohomology using only standard textbook facts (cited), with no appeal to any repository computation.
- **PROOF SKETCH (NAMED STANDARD RESULT):** the result is reduced to a precisely named theorem in the literature; we verify that theorem's hypotheses but do not reprove it.
- **GAP (OPEN SUB-LEMMA):** an honestly flagged open sub-lemma; we state exactly what must be proved.

Two analytic inputs are used repeatedly and cited once and for all: the Segal–Mitchison/Moore theory of the cohomology of topological groups ([4, 5, 6, 7]) and the Behrend–Xu / Tu–Xu–Laurent–Gengoux theory of S^1 -central extensions of Lie groupoids and its Morita invariance ([1, 2, 3]). Where a step depends on one of these it is marked **SKETCH** and the precise statement invoked is quoted.

1 Item A: objects and the cohomology theory

1.1 The space of MASAs as a homogeneous space

Fix $n \geq 2$ and $H = \mathbb{C}^n$ with its standard Hermitian inner product. Write $M_n(\mathbb{C}) = B(H)$.

Definition 1.1 (MASA). A *maximal abelian subalgebra* (MASA) of $M_n(\mathbb{C})$ is a unital $*$ -subalgebra $A \subseteq M_n(\mathbb{C})$ that is commutative and is not properly contained in any commutative unital $*$ -subalgebra. Let X denote the set of all MASAs of $M_n(\mathbb{C})$.

Lemma 1.2 (Structure and homogeneity of X). *Every MASA $A \subseteq M_n(\mathbb{C})$ equals the algebra of operators diagonal in some orthonormal basis; equivalently A is determined by the associated orthogonal decomposition $H = \bigoplus_{i=1}^n \ell_i$ into n pairwise orthogonal lines (its minimal projections). Consequently:*

- (i) *The conjugation action $U \cdot A := UAU^\dagger$ of $\text{U}(n)$ on X is transitive.*
- (ii) *Let $D = \{\text{diag}(d_1, \dots, d_n)\}$ be the standard diagonal MASA. Its stabilizer in $\text{U}(n)$ is the group $N(T)$ of monomial unitaries (one nonzero entry per row and column), i.e. $N(T) = T \rtimes S_n$ where $T = \text{U}(1)^n$ is the diagonal maximal torus and S_n acts by permutation matrices. This is the normalizer of T in $\text{U}(n)$.*

(iii) The centre $U(1) = \{\lambda I\}$ acts trivially, so the action factors through $PU(n) = U(n)/U(1)$, and

$$X \cong U(n)/N(T) \cong PU(n)/PN(T), \quad PN(T) := N(T)/U(1).$$

In particular X is a compact connected smooth manifold of dimension $\dim PU(n) - \dim PN(T) = (n^2 - 1) - (n - 1) = n^2 - n$.

Proof. A commutative $*$ -subalgebra of $M_n(\mathbb{C})$ is simultaneously diagonalizable by the spectral theorem; maximality forces it to contain a complete system of n rank-one orthogonal projections and hence to be exactly the diagonal algebra in the corresponding orthonormal basis. Two such algebras are conjugate by the unitary carrying one orthonormal basis to the other, giving (i). The stabilizer of D consists of unitaries permuting the coordinate lines $\{\mathbb{C}e_i\}$ up to phase, i.e. monomial unitaries; a monomial unitary is uniquely $P_\sigma \text{diag}(t_1, \dots, t_n)$ with $\sigma \in S_n$, $t_i \in U(1)$, giving $N(T) = T \rtimes S_n$ and the classical identification $N(T) = N_{U(n)}(T)$, proving (ii). The scalars λI fix every subspace, hence act trivially; the induced $PU(n)$ -action is still transitive with stabilizer $N(T)/U(1) = PN(T)$, giving (iii). Smoothness and the dimension count are standard for coset spaces of compact Lie groups. $\square$

Status: Complete proof. (Spectral theorem; elementary Lie theory of $N_{U(n)}(T)$.)

Convention 1.3 (The torus quotient). Throughout, $T' := T/U(1)$ where $U(1) \hookrightarrow T = U(1)^n$ is the diagonal $\lambda \mapsto (\lambda, \dots, \lambda)$. Then $T' \cong U(1)^{n-1}$ is a torus of rank $n - 1$, the diagonal scalars being S_n -invariant we have a split extension

$$1 \longrightarrow T' \longrightarrow PN(T) = T' \rtimes S_n \longrightarrow S_n \longrightarrow 1, \quad (1)$$

the splitting $S_n \hookrightarrow PN(T)$ being the images of the permutation matrices P_σ . The (Pontryagin) character lattice of T' is

$$\widehat{T'} = \text{Hom}_{\text{cont}}(T', U(1)) = \{a \in \mathbb{Z}^n : \sum_i a_i = 0\} = A_{n-1},$$

the root lattice of type A_{n-1} , with S_n acting by permuting coordinates.

1.2 The action groupoid, the stack, and the chosen cohomology theory

Definition 1.4 (Action groupoid and stack). For a Lie group G acting smoothly on a manifold M , the *action groupoid* $G \ltimes M$ has object manifold $M_0 = M$ and arrow manifold $M_1 = G \times M$, with $(g, x) : x \rightarrow g \cdot x$, composition $(h, g \cdot x)(g, x) = (hg, x)$, units (e, x) , and inverses $(g, x)^{-1} = (g^{-1}, g \cdot x)$. It presents the quotient differentiable stack $[M/G]$. We write

$$G := U(n) \ltimes X, \quad \bar{G} := PU(n) \ltimes X,$$

so that $[X/PU(n)]$ is the *MASA context stack*.

We must now fix, once and for all, a single cohomology theory. The requirements are: (a) degree 2 must classify *central extensions* by $U(1)$ (of the group, resp. of the groupoid), so that “the class of an extension” is literally an H^2 ; (b) it must be Morita invariant for Lie groupoids (Item D); (c) on a Lie group it must reduce to a theory in which the computations of Items C and E are the standard ones.

Definition 1.5 (The cohomology theory used, and its coefficients). Let $\underline{U(1)}$ denote the sheaf of smooth $U(1)$ -valued functions, regarded as a groupoid module with *trivial* action (all our extensions are central). For a Lie groupoid $\mathcal{G} = (\mathcal{G}_1 \rightrightarrows \mathcal{G}_0)$ we use the *Segal–Mitchison/Behrend–Xu groupoid cohomology* $H^\bullet(\mathcal{G}; \underline{U(1)})$, defined as the cohomology of the double complex obtained from the nerve $N_\bullet \mathcal{G}$ (the simplicial manifold with $N_p \mathcal{G} = \mathcal{G}_1 \times_{\mathcal{G}_0} \cdots \times_{\mathcal{G}_0} \mathcal{G}_1$) and a Segal–Mitchison soft resolution of $\underline{U(1)}$; see [1, §2–3] and [2, §2]. Coefficients are written multiplicatively. We record three properties, all standard in the cited references:

- (C1) **(Extensions.)** $H^2(\mathcal{G}; \underline{U(1)})$ is in natural bijection with the group of isomorphism classes of S^1 -central extensions (equivalently, $U(1)$ -gerbes with connective structure discarded) of $\mathcal{G}$; Baer sum corresponds to addition [1, Thm. 3.9].
- (C2) **(Group case.)** If $\mathcal{G} = G$ is a Lie group viewed as a one-object groupoid, then $H^\bullet(G; \underline{U(1)})$ equals the Segal–Mitchison smooth group cohomology $H_{\text{sm}}^\bullet(G; U(1))$ [4, 5]. If moreover G is finite (discrete), it equals ordinary group cohomology $H^\bullet(G; U(1))$ with $U(1)$ a trivial module.
- (C3) **(Morita invariance.)** A Morita equivalence of Lie groupoids induces an isomorphism on $H^\bullet(-; \underline{U(1)})$ [1, Thm. 3.9, Prop. 3.10], [2, §2], [3, Thm. 4].

Whenever “ $H^2(-; \underline{U(1)})$ ” appears below it means *this* theory (equivalently, on Lie groups, H_{sm}^2 ; on finite groups, ordinary H^2). We never switch silently.

Status: Complete proof as a definition; the three properties (C1)–(C3) are Proof sketch (named standard result) (they are the cited theorems of Behrend–Xu, Tu–Xu–Laurent-Gengoux, Segal–Mitchison, Moore, and we invoke rather than reprove them).

Remark 1.6 (Why this theory and not another). The alternatives are equivalent for our inputs but each obscures one requirement: Borel/Moore measurable cohomology $H_{\text{meas}}^\bullet(G; U(1))$ agrees with $H_{\text{sm}}^\bullet$ for the (compact, or Lie) groups here [5, 8] but has no built-in stack meaning; Segal–Borel equivariant cohomology $H_G^\bullet(X; \mathbb{Z})$ computes the *topological* invariants of $[X/G]$ but sits in degree shifted by one relative to $U(1)$ -central extensions. Definition 1.5 is the unique choice for which (a) the “class of a central extension” is directly H^2 , and (b) Morita invariance (Item D) is a cited theorem. All numerical outputs below are theory-independent because they are computed after Morita reduction to the compact Lie group $\text{PN}(T)$, where $H_{\text{sm}}^\bullet = H_{\text{meas}}^\bullet$.

2 Item B: the canonical class, its vacuity, and its nontriviality

2.1 Construction from the central extension of $\text{PU}(n)$

The central extension

$$1 \longrightarrow U(1) \longrightarrow U(n) \xrightarrow{\pi} \text{PU}(n) \longrightarrow 1 \quad (2)$$

has a class $[\pi] \in H_{\text{sm}}^2(\text{PU}(n); U(1))$ (property (C1) for the one-object groupoid $\text{PU}(n)$). Pick a Borel section $s: \text{PU}(n) \rightarrow U(n)$ of π ; the associated normalized 2-cocycle is

$$c_s(\bar{V}, \bar{U}) = s(\bar{V}) s(\bar{U}) s(\overline{VU})^{-1} \in U(1), \quad \bar{V}, \bar{U} \in \text{PU}(n).$$

Definition 2.1 (The canonical class). Let $F: \bar{G} = \text{PU}(n) \ltimes X \rightarrow \text{PU}(n)$ be the smooth functor sending the arrow $(\bar{U}, C): C \rightarrow \bar{U}C\bar{U}^\dagger$ to $\bar{U}$ (and the unique object to the unique object). The *canonical class* of the MASA context stack is

$$[\omega] := F^*[\pi] \in H^2(\bar{G}; \underline{U(1)}),$$

represented by the groupoid 2-cocycle $\omega((\bar{V}, \cdot), (\bar{U}, \cdot)) = c_s(\bar{V}, \bar{U})$, constant along X . Similarly define $F_0: G = \mathrm{U}(n) \ltimes X \rightarrow \mathrm{U}(n)$, and note $F = \bar{\pi} \circ F_0$ where $\bar{\pi}: \mathrm{U}(n) \rightarrow \mathrm{PU}(n)$ on labels.

2.2 Vacuity on $\mathrm{U}(n) \ltimes X$

Theorem 2.2 (Vacuity). $F_0^*[\pi] = [\omega]|_G = 0$ in $H^2(G; \underline{\mathrm{U}(1)})$, $G = \mathrm{U}(n) \ltimes X$.

Proof. On G every arrow carries an honest unitary $U \in \mathrm{U}(n)$ (not merely its class). Define a 1-cochain $\lambda: G_1 = \mathrm{U}(n) \times X \rightarrow \mathrm{U}(1)$ by

$$\lambda(U, C) := U s(\bar{U})^{-1} \in \mathrm{U}(1),$$

which lies in $\mathrm{U}(1)$ because U and $s(\bar{U})$ have the same image $\bar{U}$ in $\mathrm{PU}(n)$, so their ratio is central. The pulled-back cocycle is $\omega|_G((V, \cdot), (U, \cdot)) = s(\bar{V})s(\bar{U})s(\overline{VU})^{-1}$. Compute the groupoid coboundary of λ (trivial coefficient action):

$$(\delta\lambda)((V, \cdot), (U, \cdot)) = \lambda(V, \cdot) \lambda(U, \cdot) \lambda(VU, \cdot)^{-1} = (Vs(\bar{V})^{-1})(Us(\bar{U})^{-1})(VUs(\overline{VU})^{-1})^{-1}.$$

Since the factors are scalars they commute, and V, U cancel against $(VU)^{-1}$:

$$(\delta\lambda) = s(\bar{V})^{-1}s(\bar{U})^{-1}s(\overline{VU}) = (s(\bar{V})s(\bar{U})s(\overline{VU})^{-1})^{-1} = \omega|_G^{-1}.$$

Hence $\omega|_G = \delta(\lambda^{-1})$ is a coboundary and $[\omega]|_G = 0$. Equivalently $(U, C) \mapsto (\lambda(U, C), (U, C))$ is a strict homomorphic section of the pulled-back central extension groupoid. $\square$

Status: Complete proof. The mechanism is precisely that F_0 factors the classifying functor through $\mathrm{U}(n)$, along which (2) is (tautologically) split by the identity; the extra arrow data of the groupoid supplies the honest lift U .

Remark 2.3 (The cocycle is not itself symmetric). The vanishing is a statement about the *groupoid* class, not about c_s as a function on $\mathrm{PU}(n) \times \mathrm{PU}(n)$: for generic s the cocycle c_s disagrees with its transpose and is not a group coboundary. Theorem 2.2 says only that on $\mathrm{U}(n) \ltimes X$ the class dies; the construction must therefore be run on $\bar{G} = \mathrm{PU}(n) \ltimes X$, over which the centre acts trivially and F does not factor through $\mathrm{U}(n)$.

2.3 Nontriviality on $\mathrm{PU}(n) \ltimes X$

Theorem 2.4 (Nontriviality). *For every $n \geq 2$, $[\omega] \neq 0$ in $H^2(\bar{G}; \underline{\mathrm{U}(1)})$; indeed $[\omega]$ has order exactly n .*

Proof. By the Morita reduction (Proposition 3.2, Item D) there is an isomorphism $H^\bullet(\bar{G}; \underline{\mathrm{U}(1)}) \cong H^\bullet(\mathrm{PN}(T); \underline{\mathrm{U}(1)})$ carrying $[\omega]$ to the class $[\alpha]$ of the extension (3). By the order theorem (Theorem 4.7, Item C) $[\alpha]$ has order exactly $n \geq 2$, hence is nonzero, hence so is $[\omega]$. $\square$

Status: Complete proof given Items C and D (on which it depends and which are proved below without circularity: Item D is a Morita statement independent of the order, Item C is a finite-group computation).

3 Item D: the Morita reduction, with hypotheses verified

Theorem 3.1 (Named theorem: Morita equivalence of a transitive action groupoid with a stabilizer). *Let a Lie group G act smoothly and transitively on a manifold M , and let $H = G_{x_0}$ be the (closed) stabilizer of a point $x_0 \in M$, so $M \cong G/H$. Then the action groupoid $G \ltimes M$ is Morita equivalent to H (a one-object groupoid), with principal bibundle G itself. Consequently $H^\bullet(G \ltimes M; \underline{U}(1)) \cong H^\bullet(H; \underline{U}(1))$.*

This is standard [12, Ch. 5], [13, §2.4]; the cohomological consequence is (C3). We must verify its hypotheses *for our data* and identify the class.

Proposition 3.2 (Morita reduction of the canonical class). *The bibundle $P := \text{PU}(n)$ realises a Morita equivalence $\bar{G} = \text{PU}(n) \ltimes X \simeq_{\text{Mor}} \text{PN}(T)$, and under the induced isomorphism $H^\bullet(\bar{G}; \underline{U}(1)) \cong H^\bullet(\text{PN}(T); \underline{U}(1))$ the canonical class $[\omega]$ corresponds to the class $[\alpha]$ of the central extension*

$$1 \longrightarrow \text{U}(1) \longrightarrow N(T) \longrightarrow \text{PN}(T) \longrightarrow 1. \quad (3)$$

Proof. Hypotheses of Theorem 3.1. By Lemma 1.2 the $\text{PU}(n)$ -action on X is smooth and transitive with closed stabilizer $\text{PN}(T)$ at the base $\text{MASA } D$, and $X \cong \text{PU}(n)/\text{PN}(T)$. All groups are compact Lie, so the stabilizer is closed and the quotient map is a smooth principal $\text{PN}(T)$ -bundle. Hence Theorem 3.1 applies with $G = \text{PU}(n)$, $M = X$, $H = \text{PN}(T)$.

The bibundle, explicitly. Let $P = \text{PU}(n)$ with: (i) a right $\text{PN}(T)$ -action by right translation $p \cdot h = ph$, which is free and proper (right translation in a compact Lie group by a closed subgroup) with quotient $P/\text{PN}(T) = X$ and moment map $\tau: P \rightarrow X$, $\tau(p) = p \cdot D := pDp^\dagger$; (ii) a left $\bar{G}$ -action over τ : an arrow $(\bar{U}, x)$ with $x = \tau(p)$ acts by $(\bar{U}, x) \cdot p = \bar{U}p$; this is well defined since $\tau(\bar{U}p) = \bar{U} \cdot \tau(p)$, and it is *principal*, i.e. τ is a surjective submersion and $\bar{G}$ acts freely and transitively on the τ -fibres (given p, p' with $\tau(p) = \tau(p')$ the unique arrow is $(p'p^{-1}, \tau(p))$). The two actions commute: $(\bar{U}p)h = \bar{U}(ph)$. Thus P is a $\bar{G}$ - $\text{PN}(T)$ principal bibundle, which is the definition of a Morita equivalence [12, §5.4]. Property (C3) then gives the cohomology isomorphism.

Transport of the class. The classifying functor $F: \bar{G} \rightarrow \text{PU}(n)$ of Definition 2.1 restricts, under the equivalence $\bar{G} \simeq_{\text{Mor}} \text{PN}(T)$, to the inclusion of isotropy $\iota: \text{PN}(T) \hookrightarrow \text{PU}(n)$ (an arrow of $\bar{G}$ fixing D has label in the stabilizer $\text{PN}(T)$, and F reads off that label). Naturality of pullback gives $[\omega] = F^*[\pi] \mapsto \iota^*[\pi]$ under the isomorphism. Finally $\iota^*[\pi]$ is the class of the pullback of (2) along $\iota: \text{PN}(T) \hookrightarrow \text{PU}(n)$, namely of

$$1 \rightarrow \text{U}(1) \rightarrow \pi^{-1}(\text{PN}(T)) \rightarrow \text{PN}(T) \rightarrow 1.$$

Because $\text{U}(1) = \ker \pi \subseteq N(T)$ (scalars are monomial) and $N(T)/\text{U}(1) = \text{PN}(T)$, we have $\pi^{-1}(\text{PN}(T)) = N(T)$, so the pullback is exactly (3). Hence $[\omega] \leftrightarrow [\alpha]$. $\square$

Status: Complete proof for the bibundle construction and the transport-of-class identification; Proof sketch (named standard result) for the single invocation of (C3) (the Morita invariance theorem [1, 2, 3]), whose hypotheses (both actions principal, actions commute) are verified above. No step here uses “by Morita invariance” without exhibiting the principal bibundle and checking its axioms.

Remark 3.3 (On the naturality step). The only non-mechanical point is that F corresponds to ι under the equivalence. This is because a Morita equivalence $\bar{G} \simeq H$ identifies the category of $\bar{G}$ -representations/cocycles with that of H , and the isotropy inclusion $H \hookrightarrow \bar{G}$ (as the automorphisms of the base object) composed with F is ι . For the H^2 -level statement (C1) this is the assertion that the S^1 -central extension classified by $[\omega]$ restricts on isotropy to (3); see [1, Prop. 3.10].

4 Item C: the order is exactly n

By Proposition 3.2 it suffices to compute the order of $[\alpha] \in H^2(\text{PN}(T); \text{U}(1))$, the class of (3). We prove $n[\alpha] = 0$ and $k[\alpha] \neq 0$ for $1 \leq k < n$.

4.1 Upper bound: $n[\alpha] = 0$ via the determinant twist

Lemma 4.1 (Determinant on $N(T)$). $\det: N(T) \rightarrow \text{U}(1)$, $\det(P_\sigma \text{diag}(t_1, \dots, t_n)) = \text{sgn}(\sigma) \prod_i t_i$, is a continuous homomorphism with $\det(\lambda g) = \lambda^n \det(g)$ for $\lambda \in \text{U}(1)$, $g \in N(T)$.

Proof. The determinant is multiplicative and continuous on $\text{U}(n) \supseteq N(T)$; homogeneity is $\det(\lambda I \cdot g) = \lambda^n \det g$. $\square$

Status: Complete proof.

Theorem 4.2 (Upper bound). $n[\alpha] = 0$ in $H^2(\text{PN}(T); \text{U}(1))$.

Proof. The class $n[\alpha]$ is (property (C1), Baer sum) represented by the pushout of (3) along the n -th power map $[n]: \text{U}(1) \rightarrow \text{U}(1)$, $z \mapsto z^n$. Concretely,

$$E_n = (\text{U}(1) \times N(T)) / \sim, \quad (z\lambda^{-n}, \lambda g) \sim (z, g) \quad (\lambda \in \text{U}(1)),$$

where $\text{U}(1) \hookrightarrow N(T)$ as scalars; write $[z, g]$ for classes. The quotient map $E_n \rightarrow \text{PN}(T)$, $[z, g] \mapsto \bar{g}$, is a central extension with kernel $\{[z, I] : z \in \text{U}(1)\} \cong \text{U}(1)$, and its class is $n[\alpha]$. Define

$$\sigma: \text{PN}(T) \longrightarrow E_n, \quad \sigma(\bar{g}) = [(\det g)^{-1}, g].$$

Well defined: if $g' = \lambda g$ ($\lambda \in \text{U}(1)$) then, using Lemma 4.1 and the relation,

$$[(\det g')^{-1}, g'] = [(\lambda^n \det g)^{-1}, \lambda g] = [\lambda^{-n} (\det g)^{-1}, \lambda g] = [(\det g)^{-1}, g].$$

Homomorphism: the product in E_n is $[z, g][w, h] = [zw, gh]$ (the $\text{U}(1)$ -factor is central), so with $\det$ multiplicative

$$\sigma(\bar{g})\sigma(\bar{h}) = [(\det g)^{-1}(\det h)^{-1}, gh] = [(\det gh)^{-1}, gh] = \sigma(\overline{gh}).$$

Thus σ is a continuous homomorphic splitting of $E_n \rightarrow \text{PN}(T)$, so $n[\alpha] = 0$. $\square$

Status: Complete proof. The section is genuinely a Lie-group homomorphism (no measurability subtlety), being polynomial in matrix entries. This is the group-level shadow of the n -torsion of Dixmier–Douady classes of $\text{PU}(n)$ -bundles [16], but the proof above is self-contained.

4.2 Lower bound: $k[\alpha] \neq 0$ for $1 \leq k < n$

The lower bound is proved by restricting to a *finite* subgroup, where no continuity or measurability subtlety can arise and all homomorphisms to $\text{U}(1)$ are classical characters.

Definition 4.3 (Finite scalar shadow). Let $\mu_n = \{z \in \text{U}(1) : z^n = 1\}$, $D = \mu_n^n \subset T$ (diagonal matrices with entries in μ_n), and $Z = \{(\zeta, \dots, \zeta) : \zeta \in \mu_n\} = D \cap \text{U}(1) \cong \mu_n$ the scalar subgroup. Put $W_n = D \rtimes S_n = \mu_n \wr S_n \subset N(T)$ and $\overline{W}_n = W_n/Z = (D/Z) \rtimes S_n \subset \text{PN}(T)$.

Lemma 4.4 (Restriction of $[\alpha]$ to $\overline{W}_n$). *The restriction $[\alpha]|_{\overline{W}_n} \in H^2(\overline{W}_n; \text{U}(1))$ is the image, under $\mu_n \hookrightarrow \text{U}(1)$, of the class of the finite central extension*

$$1 \longrightarrow \mu_n \longrightarrow W_n \longrightarrow \overline{W}_n \longrightarrow 1. \quad (4)$$

Proof. The preimage of $\overline{W}_n$ under $N(T) \rightarrow \text{PN}(T)$ is $U(1) \cdot W_n$, and $U(1) \cdot W_n \cong (U(1) \times W_n)/\{(\zeta^{-1}, \zeta) : \zeta \in \mu_n = U(1) \cap W_n\}$ is exactly the pushout of (4) along $\mu_n \hookrightarrow U(1)$. Restriction of extension classes is restriction of the group, giving the claim. $\square$

Status: Complete proof.

Lemma 4.5 (S_n -equivariant homomorphism lemma). *Let S_n act on μ_n^n (equivalently on $\mathbb{Z}_n^n$) by permuting coordinates and trivially on $U(1)$. Every S_n -invariant character $\rho: \mu_n^n \rightarrow U(1)$ has the form*

$$\rho(\zeta_n^{a_1}, \dots, \zeta_n^{a_n}) = \zeta_n^{k \sum_i a_i}, \quad k \in \mathbb{Z}_n,$$

($\zeta_n = e^{2\pi i/n}$); there are exactly n of them, and each satisfies $\rho(\zeta_n, \dots, \zeta_n) = \zeta_n^{kn} = 1$. Equivalently, the only S_n -equivariant homomorphisms $\mathbb{Z}_n^n \rightarrow \mathbb{Z}_n$ are $x \mapsto k \sum_i x_i$, and all annihilate the diagonal generator.

Proof. A character of the finite abelian group μ_n^n is $x \mapsto \prod_i \rho(e_i)^{a_i}$ with $\rho(e_i) \in \mu_n$; write $\rho(e_i) = \zeta_n^{c_i}$. Since S_n permutes the standard generators e_i transitively, invariance $\rho(\sigma \cdot x) = \rho(x)$ forces $c_{\sigma(i)} = c_i$ for all σ , hence $c_1 = \dots = c_n =: k$. Then $\rho(x) = \zeta_n^{k \sum a_i}$ and on the diagonal $\sum a_i = n$, so $\rho = \zeta_n^{kn} = 1$. There are exactly n choices $k \in \mathbb{Z}_n$. $\square$

Status: Complete proof. This is the exact point at which “ S_n -equivariance” enters: it forces the invariant character to be a multiple of the coordinate sum, which is $\equiv 0$ on the diagonal modulo n .

Theorem 4.6 (Lower bound). *For every integer k with $1 \leq k < n$, $k[\alpha] \neq 0$ in $H^2(\text{PN}(T); U(1))$.*

Proof. Restriction $H^2(\text{PN}(T); U(1)) \rightarrow H^2(\overline{W}_n; U(1))$ is a homomorphism, so it suffices to show $k[\alpha]|_{\overline{W}_n} \neq 0$ for $1 \leq k < n$. By Lemma 4.4, $k[\alpha]|_{\overline{W}_n}$ is the class of the pushout of (4) along $\mu_n \xrightarrow{z \mapsto z^k} U(1)$. This pushout *splits* (i.e. the class vanishes) if and only if there is a homomorphism $\rho: W_n \rightarrow U(1)$ extending $\zeta \mapsto \zeta^k$ on $\mu_n = Z$, i.e. with

$$\rho(\zeta_n I) = \zeta_n^k = e^{2\pi i k/n} (\neq 1 \text{ since } 1 \leq k < n). \quad (5)$$

Now for *any* homomorphism $\rho: W_n = D \rtimes S_n \rightarrow U(1)$, the restriction $\rho|_D$ is S_n -invariant: for $s \in S_n \subset W_n$ and $d \in D$ we have $sds^{-1} = s \cdot d \in D$ and, $U(1)$ being abelian, $\rho(s \cdot d) = \rho(sds^{-1}) = \rho(d)$. By Lemma 4.5, $\rho|_D(\zeta_n I) = \rho(\zeta_n, \dots, \zeta_n) = 1$, contradicting (5). Hence no such ρ exists, so $k[\alpha]|_{\overline{W}_n} \neq 0$, so $k[\alpha] \neq 0$. $\square$

Status: Complete proof.

Corollary 4.7 (Order exactly n). *The canonical class $[\alpha]$ — equivalently $[\omega] \in H^2([X/\text{PU}(n)]; \underline{U(1)})$ — has order exactly n .*

Proof. $n[\alpha] = 0$ (Theorem 4.2) and $k[\alpha] \neq 0$ for $1 \leq k < n$ (Theorem 4.6). $\square$

Status: Complete proof.

4.3 Why no “exotic” splitting evades the argument

Three objections must be ruled out; all three are handled by the restriction to the finite group $\overline{W}_n$.

Proposition 4.8 (Robustness of the lower bound). *The conclusion $k[\alpha] \neq 0$ ($1 \leq k < n$) is immune to:*

- (a) **Nonlinear “splittings.”** *A splitting of a central extension of groups is by definition a group homomorphism s with $\pi \circ s = \text{id}$; “nonlinear” set-maps are irrelevant, since vanishing of the class means existence of such a homomorphism. The proof of Theorem 4.6 shows no homomorphism ρ with the required value exists.*
- (b) **Discontinuous homomorphisms.** *Over the ambient Lie group $\text{PN}(T)$ there exist, by a Hamel-basis/choice construction, discontinuous homomorphisms $\text{PN}(T) \rightarrow \text{U}(1)$. But any hypothetical splitting of $k[\alpha]$ over $\text{PN}(T)$ restricts to a splitting over the finite subgroup $\overline{W}_n$, where every homomorphism is automatically continuous and is a classical character. Theorem 4.6 rules those out. Formally: restriction $H^2(\text{PN}(T); \text{U}(1)) \rightarrow H^2(\overline{W}_n; \text{U}(1))$ is a homomorphism, and $k[\alpha] \mapsto k[\alpha]|_{\overline{W}_n} \neq 0$, so $k[\alpha] \neq 0$ regardless of the regularity of any putative section upstairs.*
- (c) **Non-module (twisted) splittings.** *The coefficient module is central ($\text{U}(1)$ with trivial action); a section respecting only some twisted action is not a splitting of (3). Since (3) is central, the notion of splitting is unambiguous and is the one used.*

Proof. Immediate from the definitions and from the naturality of restriction, as spelled out in (a)–(c). $\square$

Status: Complete proof.

Remark 4.9 (Where the upper and lower bounds meet). The upper bound used a specific, everywhere-defined homomorphic section (the determinant twist), so it is unconditional. The lower bound is a nonexistence statement pushed down to a finite group, so it too is unconditional. Together they pin the order at exactly n with no appeal to any computer search; the repository’s `thmG_general.py` merely double-checks Lemma 4.5 and the order over $n = 2, \dots, 8$.

5 Item E: $H^2(\text{PN}(T); \text{U}(1)) \cong \mathbb{Z}/n \oplus \mathbb{Z}_2$

We now compute the full group, using the Lyndon–Hochschild–Serre (LHS) spectral sequence of the split extension (1). All coefficients are $\text{U}(1)$ with trivial action (Definition 1.5); the outer group S_n is finite, so $H^\bullet(S_n; -)$ is ordinary group cohomology, while the inner torus contributes smooth (Segal–Mitchison) cohomology.

5.1 The spectral sequence and its coefficient inputs

Theorem 5.1 (Existence of the LHS spectral sequence). *For the split extension (1) there is a first-quadrant cohomological spectral sequence*

$$E_2^{p,q} = H^p(S_n; H_{\text{sm}}^q(T'; \text{U}(1))) \implies H_{\text{sm}}^{p+q}(\text{PN}(T); \text{U}(1)),$$

with the usual multiplicative and edge structure, converging to a finite filtration of the abutment.

Status: Proof sketch (named standard result). This is the Hochschild–Serre spectral sequence for measurable/smooth cohomology of the extension of locally compact groups $T' \rightarrow \text{PN}(T) \rightarrow S_n$, due to Moore [6, §§2–3] (see also [10, 11] for the topological/site formulation); the inner terms $H_{\text{sm}}^q(T'; \text{U}(1))$ carry the S_n -action induced by conjugation, and because S_n is finite and discrete the outer cohomology is ordinary group cohomology. We invoke Moore’s theorem; its hypotheses (a closed normal subgroup of a second countable locally compact group) hold.

We now identify each relevant coefficient group.

Lemma 5.2 (Torus coefficients). *As S_n -modules,*

$$H_{\text{sm}}^0(T'; \text{U}(1)) = \text{U}(1) \text{ (trivial)}, \quad H_{\text{sm}}^1(T'; \text{U}(1)) = \widehat{T'} = A_{n-1}, \quad H_{\text{sm}}^2(T'; \text{U}(1)) = 0.$$

Proof. Degree 0 is the constants. Degree 1: $H_{\text{sm}}^1(T'; \text{U}(1)) = \text{Hom}_{\text{cont}}(T', \text{U}(1)) = \widehat{T'} = A_{n-1}$ (Convention 1.3), with the permutation S_n -action.

Degree 2: We claim $H_{\text{sm}}^2(T'; \text{U}(1)) = 0$, i.e. every smooth $\text{U}(1)$ -central extension of the torus T' splits. Use the exponential sequence of trivial T' -modules $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{R} \xrightarrow{\exp(2\pi i \cdot)} \text{U}(1) \rightarrow 0$ and its long exact sequence in $H_{\text{sm}}^\bullet(T'; -)$. For the compact group T' , continuous cohomology with $\mathbb{R}$ coefficients vanishes in positive degree, $H_{\text{sm}}^k(T'; \mathbb{R}) = 0$ for $k \geq 1$ (invariant integration/averaging [9, Ch. IX], [5]), so the connecting maps give $H_{\text{sm}}^k(T'; \text{U}(1)) \cong H_{\text{sm}}^{k+1}(T'; \mathbb{Z})$ for $k \geq 1$. Finally, for a compact Lie group G , $H_{\text{sm}}^k(G; \mathbb{Z}) \cong H^k(BG; \mathbb{Z})$ (Segal [4], Wigner [7]); for $G = T' \cong \text{U}(1)^{n-1}$, $BG = (\mathbb{C}P^\infty)^{n-1}$ has $H^\bullet(BG; \mathbb{Z}) = \mathbb{Z}[x_1, \dots, x_{n-1}]$ with $\deg x_i = 2$, concentrated in even degrees and torsion-free. Hence $H^3(BT'; \mathbb{Z}) = 0$, so $H_{\text{sm}}^2(T'; \text{U}(1)) \cong H_{\text{sm}}^3(T'; \mathbb{Z}) \cong H^3(BT'; \mathbb{Z}) = 0$. $\square$

Status: Proof sketch (named standard result). The vanishing $H_{\text{sm}}^2(T'; \text{U}(1)) = 0$ is reduced to the two named results $H_{\text{sm}}^k(\text{compact}; \mathbb{R}) = 0$ ($k \geq 1$) and $H_{\text{sm}}^\bullet(G; \mathbb{Z}) \cong H^\bullet(BG; \mathbb{Z})$ (Segal–Wigner), whose hypotheses (compactness, Lie) hold. Intuitively: a continuous bicharacter $T' \times T' \rightarrow \text{U}(1)$ is trivial (each slice is a character, hence locally constant integer-valued), so the extension is abelian, and abelian smooth extensions of a torus by $\text{U}(1)$ split by the BG -computation.

Lemma 5.3 (S_n -cohomology inputs). *With trivial $\text{U}(1)$ -coefficients and the permutation module A_{n-1} :*

- (i) $H^0(S_n; A_{n-1}) = (A_{n-1})^{S_n} = 0$.
- (ii) $H^1(S_n; A_{n-1}) \cong \mathbb{Z}/n$.
- (iii) $H^2(S_n; \text{U}(1)) \cong M(S_n)$, the Schur multiplier: $\mathbb{Z}_2$ for $n \geq 4$ and 0 for $n \leq 3$.

Proof. (i) An S_n -invariant vector of $\mathbb{Z}^n$ is a scalar multiple of $(1, \dots, 1)$; requiring coordinate sum 0 forces it to be 0.

(ii) Use the short exact sequence of S_n -modules $0 \rightarrow A_{n-1} \rightarrow \mathbb{Z}^n \xrightarrow{\Sigma} \mathbb{Z} \rightarrow 0$, where $\mathbb{Z}^n$ is the permutation module (i.e. $\mathbb{Z}[S_n/S_{n-1}]$) and $\mathbb{Z}$ is trivial. Its long exact sequence reads

$$H^0(S_n; \mathbb{Z}^n) \xrightarrow{\Sigma_*} H^0(S_n; \mathbb{Z}) \xrightarrow{\delta} H^1(S_n; A_{n-1}) \longrightarrow H^1(S_n; \mathbb{Z}^n).$$

By Shapiro’s lemma $H^\bullet(S_n; \mathbb{Z}^n) = H^\bullet(S_n; \mathbb{Z}[S_n/S_{n-1}]) = H^\bullet(S_{n-1}; \mathbb{Z})$, so $H^0(S_n; \mathbb{Z}^n) = \mathbb{Z} \cdot (1, \dots, 1)$ and $H^1(S_n; \mathbb{Z}^n) = H^1(S_{n-1}; \mathbb{Z}) = \text{Hom}(S_{n-1}, \mathbb{Z}) = 0$. The map Σ_* sends $(1, \dots, 1)$ to $n \in \mathbb{Z} = H^0(S_n; \mathbb{Z})$, i.e. is multiplication by n ; hence $H^1(S_n; A_{n-1}) \cong \text{coker}(\mathbb{Z} \xrightarrow{n} \mathbb{Z}) = \mathbb{Z}/n$.

(iii) For finite G and divisible $\text{U}(1)$, the universal-coefficient/Hopf formula gives $H^2(G; \text{U}(1)) \cong \text{Hom}(H_2(G; \mathbb{Z}), \text{U}(1)) \cong H_2(G; \mathbb{Z}) = M(G)$ (the last isomorphism because $H_2(G; \mathbb{Z})$ is finite abelian, hence non-canonically self-Pontryagin-dual, and $\text{Ext}(H_1(G), \text{U}(1)) = 0$ as $\text{U}(1)$ is injective). Schur’s computation of the multiplier of the symmetric group is $M(S_n) = \mathbb{Z}_2$ for $n \geq 4$ and 0 for $n \in \{2, 3\}$ [14, Ch. 2], [15]. $\square$

Status: (i),(ii) Complete proof; (iii) Proof sketch (named standard result) (Schur's multiplier of S_n , cited).

5.2 The E_2 page in total degree 2, and all differentials

Assembling Lemmas 5.2–5.3, the anti-diagonal $p + q = 2$ of the E_2 page is

$$E_2^{0,2} = H^0(S_n; H_{\text{sm}}^2(T'; U(1))) = H^0(S_n; 0) = 0, \quad E_2^{1,1} = H^1(S_n; A_{n-1}) = \mathbb{Z}/n,$$

$$E_2^{2,0} = H^2(S_n; U(1)) = \begin{cases} \mathbb{Z}_2, & n \geq 4, \\ 0, & n \leq 3, \end{cases}$$

and, in the neighbouring spots we shall need,

$$E_2^{0,1} = H^0(S_n; A_{n-1}) = 0, \quad E_2^{3,0} = H^3(S_n; U(1)).$$

Proposition 5.4 (All differentials touching total degree 2 vanish). *In the spectral sequence of Theorem 5.1:*

- (i) every differential into the bottom row $E_r^{\bullet,0}$ vanishes; in particular $E_\infty^{p,0} = E_2^{p,0} = H^p(S_n; U(1))$ for all p ;
- (ii) consequently $d_2: E_2^{1,1} \rightarrow E_2^{3,0}$ is zero, and since no differential enters $E^{1,1}$, we get $E_\infty^{1,1} = E_2^{1,1} = \mathbb{Z}/n$;
- (iii) $E_\infty^{0,2} = E_2^{0,2} = 0$ and $E_\infty^{2,0} = E_2^{2,0}$.

Proof. (i) The extension (1) is *split*: the section $j: S_n \hookrightarrow \text{PN}(T)$ (permutation matrices) satisfies $p \circ j = \text{id}_{S_n}$ where $p: \text{PN}(T) \rightarrow S_n$. With trivial coefficients $U(1)$, inflation $\text{inf} = p^*: H^\bullet(S_n; U(1)) \rightarrow H^\bullet(\text{PN}(T); U(1))$ and restriction $\text{res} = j^*$ satisfy $\text{res} \circ \text{inf} = (p \circ j)^* = \text{id}$. Hence inflation is (split) injective in every degree. The image of inflation is precisely the bottom filtration piece $F^p H^p = E_\infty^{p,0}$, and the edge map $E_2^{p,0} \twoheadrightarrow E_\infty^{p,0}$ composed with $E_\infty^{p,0} \hookrightarrow H^p$ is inflation. Injectivity of inflation forces the surjection $E_2^{p,0} \twoheadrightarrow E_\infty^{p,0}$ to be an isomorphism, so no differential may hit the bottom row.

(ii) The only differential into $E^{3,0}$ that could reduce it is $d_2: E_2^{1,1} \rightarrow E_2^{3,0}$ (and $d_3: E_3^{0,2} \rightarrow E_3^{3,0}$); by (i) both vanish, in particular $d_2|_{E_2^{1,1}} = 0$. No differential enters $E^{1,1}$ (sources $E_r^{1-r,r}$ vanish for $r \geq 2$ as $1 - r < 0$), and the only possible outgoing differential $d_2: E_2^{1,1} \rightarrow E_2^{3,0}$ is zero; hence $E_\infty^{1,1} = E_2^{1,1} = \mathbb{Z}/n$.

(iii) $E_2^{0,2} = 0$ has nothing in it; and the only differential into $E^{2,0}$ is $d_2: E_2^{0,1} \rightarrow E_2^{2,0}$ with $E_2^{0,1} = 0$, so $E_\infty^{2,0} = E_2^{2,0}$ (also a special case of (i)). $\square$

Status: Complete proof. The vanishing of the crucial differential $d_2|_{E^{1,1}}$ is forced by the *splitting* of (1) alone; it does *not* require the order theorem.

5.3 The extension problem, and the answer

Theorem 5.5 (Full second cohomology). *For all $n \geq 2$,*

$$H^2(\text{PN}(T); U(1)) \cong \mathbb{Z}/n \oplus H^2(S_n; U(1)) = \begin{cases} \mathbb{Z}/n \oplus \mathbb{Z}_2, & n \geq 4, \\ \mathbb{Z}/3, & n = 3, \\ \mathbb{Z}/2, & n = 2, \end{cases}$$

and the canonical class $[\alpha]$ maps to a generator of the $\mathbb{Z}/n$ summand.

Proof. By Proposition 5.4 the degree-2 filtration of $H^2 := H^2(\text{PN}(T); \mathbb{U}(1))$ is

$$0 = F^3 \subseteq F^2 \subseteq F^1 = F^0 = H^2, \quad F^2 = E_\infty^{2,0} = H^2(S_n; \mathbb{U}(1)), \quad F^1/F^2 = E_\infty^{1,1} = \mathbb{Z}/n, \quad E_\infty^{0,2} = 0,$$

i.e. a short exact sequence

$$0 \longrightarrow H^2(S_n; \mathbb{U}(1)) \xrightarrow{\text{inf}} H^2 \xrightarrow{\rho} \mathbb{Z}/n \longrightarrow 0. \quad (6)$$

The sequence splits, for every n . The sub $F^2 = \text{im}(\text{inf})$ is the image of inflation, and by Proposition 5.4(i) the restriction $\text{res}: H^2 \rightarrow H^2(S_n; \mathbb{U}(1))$ satisfies $\text{res} \circ \text{inf} = \text{id}$. Thus res is a retraction of the inclusion inf in (6), so

$$H^2 = \text{im}(\text{inf}) \oplus \ker(\text{res}) \cong H^2(S_n; \mathbb{U}(1)) \oplus (H^2/F^2) = H^2(S_n; \mathbb{U}(1)) \oplus \mathbb{Z}/n.$$

This gives the stated groups using Lemma 5.3(iii) and the low- n values.

$[\alpha]$ *generates the $\mathbb{Z}/n$.* By Corollary 4.7, $[\alpha]$ has order exactly n . Its image $\rho([\alpha]) \in \mathbb{Z}/n$ has order $n/|\langle [\alpha] \rangle \cap F^2|$; since $F^2 = H^2(S_n; \mathbb{U}(1))$ has order dividing 2, and $\ker(\text{res})$ is a complement with $\text{res}([\alpha]) \in H^2(S_n; \mathbb{U}(1))$, we argue directly: the projection $q := (H^2 \rightarrow \ker(\text{res}) \cong \mathbb{Z}/n)$ along F^2 agrees with ρ up to the chosen splitting, and $q([\alpha])$ has order equal to that of $[\alpha]$ minus its F^2 -part. Concretely, write $[\alpha] = (u, v) \in F^2 \oplus \mathbb{Z}/n$. Then $\text{res}([\alpha]) = u$, and by Proposition 4.8 applied through the finite shadow, $k[\alpha] \neq 0$ for $1 \leq k < n$; if v had order $d < n$ in $\mathbb{Z}/n$ then $d[\alpha] = (du, 0) = (du, 0)$ would lie in F^2 , of order ≤ 2 , forcing $2d[\alpha] = 0$, i.e. $n \mid 2d$. Combined with $d \mid n$ and $d < n$ this leaves only n even and $d = n/2$. That residual case is excluded in Proposition 5.6 below; granting it, v has order n , i.e. generates. $\square$

Status: Complete proof for the group isomorphism $H^2 \cong \mathbb{Z}/n \oplus H^2(S_n; \mathbb{U}(1))$ (the splitting of (6) is unconditional, via the retraction res); the identification of $[\alpha]$ as a generator of the $\mathbb{Z}/n$ summand is Complete proof for n odd and rests on Proposition 5.6 for n even.

Proposition 5.6 ($[\alpha]$ generates modulo the Schur summand). $\rho([\alpha])$ generates $E_\infty^{1,1} = \mathbb{Z}/n$.

- For n odd this is COMPLETE PROOF: $|\langle [\alpha] \rangle| = n$ is coprime to $|F^2| \in \{1, 2\}$, so $\langle [\alpha] \rangle \cap F^2 = 0$ and $\rho([\alpha])$ has order n .
- For n even this is a GAP (OPEN SUB-LEMMA): one must exclude $(n/2)[\alpha] \in F^2$, i.e. that the order-2 element $(n/2)[\alpha]$ is inflated from the Schur class of S_n .

Status: n odd: Complete proof. n even: Gap (open sub-lemma) (stated precisely below).

Remark 5.7 (Exact statement of the residual gap, and partial evidence). The residual open sub-lemma is:

For n even, $(n/2)[\alpha] \notin \text{im}(\text{inf}: H^2(S_n; \mathbb{U}(1)) \rightarrow H^2(\text{PN}(T); \mathbb{U}(1)))$; equivalently, $\text{res}_{S_n}((n/2)[\alpha]) = 0$ does not coincide with $(n/2)[\alpha]$ being the inflated Schur class.

Two remarks bound the stakes. (1) The isomorphism type $H^2 \cong \mathbb{Z}/n \oplus \mathbb{Z}_2$ is *already proved unconditionally* (Theorem 5.5); the gap concerns only whether the *specific* class $[\alpha]$ is a generator of the $\mathbb{Z}/n$ factor versus lying in $\mathbb{Z}/n \oplus \mathbb{Z}_2$ with a nonzero $\mathbb{Z}_2$ -component (which would still leave $[\alpha]$ of order n and the group unchanged). (2) The natural attempt — detect a nonzero image on the abelian part $D/Z \subset \overline{W}_n$ — fails because $[\alpha]|_{D/Z} = 0$: the finite abelian extension $1 \rightarrow Z \rightarrow$

$D \rightarrow D/Z \rightarrow 1$ splits (the retraction $D = \mu_n^n \rightarrow \mu_n$, $x \mapsto x_1$, restricts to the identity on Z), so $[\alpha]$ is invisible on D/Z and lives genuinely in the mixed $E^{1,1}$ -stratum. Closing the gap therefore requires computing the $U(1)$ -valued 2-cocycle of $\overline{W}_n$ on a permutation $\times$ sign pair (the transgression $S_n \rightarrow \widehat{T'}$), or equivalently identifying $\text{res}_{S_n}(n/2)[\alpha]$ with a 2-cocycle of S_n and checking it is a coboundary. A direct computation of the $E^{1,1}$ -transgression cocycle $\gamma(\sigma) = e_{\sigma^{-1}(n)} - e_n \in A_{n-1}$ for the section of Convention 1.3 strongly indicates γ is a *generator* of $H^1(S_n; A_{n-1}) = \mathbb{Z}/n$ (it is the connecting image of $1 \in \mathbb{Z}$ under $\Sigma: \mathbb{Z}^n \rightarrow \mathbb{Z}$), which would force $\rho([\alpha])$ to generate and close the gap; we do not certify the cocycle-identity bookkeeping here and so leave it flagged.

5.4 Coefficient conventions, topology assumptions, low- n exceptions

- **Coefficients.** Everywhere $U(1)$ is a *trivial* module (central extensions); $A_{n-1} = \widehat{T'}$ carries the permutation S_n -action; $\mathbb{Z}$ and $\mathbb{R}$ in Lemma 5.2 are trivial modules.
- **Topology.** T' , $PN(T)$, $PU(n)$, $U(n)$ are compact Lie groups; “ H_{sm}^2 ” is Segal–Mitchison smooth cohomology, which for these (second countable, compact) groups coincides with Moore measurable cohomology [5, 8]. The inner cohomology $H_{\text{sm}}^\bullet(T'; U(1))$ is the smooth one; the outer $H^\bullet(S_n; -)$ is ordinary (discrete) group cohomology because S_n is finite. The identification $H_{\text{sm}}^\bullet(T'; \mathbb{Z}) \cong H^\bullet(BT'; \mathbb{Z})$ requires compact Lie T' (Segal–Wigner).
- **Low- n exceptions.** $n = 2$: $S_2 = \mathbb{Z}_2$, $A_1 = \mathbb{Z}$ with the sign action, $H^1(\mathbb{Z}_2; \mathbb{Z}_{\text{sgn}}) = \mathbb{Z}/2$ and $H^2(\mathbb{Z}_2; U(1)) = 0$, so $H^2(PN(T); U(1)) = \mathbb{Z}/2$ (no Schur factor). $n = 3$: $M(S_3) = 0$, so $H^2(PN(T); U(1)) = \mathbb{Z}/3$. The $\mathbb{Z}_2$ summand appears only for $n \geq 4$, exactly the range in which the Schur multiplier is nontrivial and (Paper A) the Peres–Mermin shadow is available.

Status: Complete proof (bookkeeping of conventions and the low- n computations).

6 Status of each step

The table records, for every lettered item and every numbered result, whether the argument here is a *complete proof* (checkable without repository scripts), a *proof sketch* reduced to a named standard theorem (with hypotheses verified), or an explicit *gap*.

Item / Result	Content	Status
A	Objects; $X \cong \mathrm{PU}(n)/\mathrm{PN}(T)$; choice of cohomology theory	Complete (defs); Sketch for (C1)–(C3) [BX, TXL, Seg, Moore]
Lem. 1.2	Structure/homogeneity of X	Complete
Def. 1.5	Groupoid cohomology $H^\bullet(-; \underline{\mathrm{U}(1)})$	Complete def.; Sketch (C1)–(C3)
B	Canonical class; vacuity; nontriviality	Complete (given C, D)
Thm. 2.2	Vacuity on $\mathrm{U}(n) \ltimes X$	Complete
Thm. 2.4	Nontriviality on $\mathrm{PU}(n) \ltimes X$	Complete given C, D
D	Morita reduction to $\mathrm{PN}(T)$	Complete bibundle; Sketch (C3)
Thm. 3.1	$G \ltimes (G/H) \simeq_{\mathrm{Mor}} H$	Sketch [MM, dH, BX]
Prop. 3.2	Bibundle $\mathrm{PU}(n)$; class $\mapsto (3)$	Complete (constr. & transport); Sketch (C3)
C	Order exactly n	Complete
Thm. 4.2	$n[\alpha] = 0$ (determinant twist)	Complete
Lem. 4.5	S_n -equivariant hom lemma	Complete
Thm. 4.6	$k[\alpha] \neq 0, 1 \leq k < n$	Complete
Cor. 4.7	Order = n	Complete
Prop. 4.8	No nonlinear/discontinuous/non-module evasion	Complete
E	$H^2(\mathrm{PN}(T); \mathrm{U}(1)) \cong \mathbb{Z}/n \oplus \mathbb{Z}_2$ ($n \geq 4$)	Complete (iso); Sketch inputs; Gap (generator, n even)
Thm. 5.1	LHS spectral sequence exists	Sketch [Moore IV]
Lem. 5.2	$H_{\mathrm{sm}}^{\leq 2}(T'; \mathrm{U}(1))$	$q=0, 1$ Complete; $q=2=0$ Sketch [Seg, Wig]
Lem. 5.3	$H^0=0, H^1=\mathbb{Z}/n, H^2(S_n; \mathrm{U}(1))$	(i),(ii) Complete; (iii) Sketch [Schur]
Prop. 5.4	All differentials to deg. 2 vanish	Complete
Thm. 5.5	$H^2 \cong \mathbb{Z}/n \oplus H^2(S_n; \mathrm{U}(1))$	Complete (group iso)
Prop. 5.6	$[\alpha]$ generates $\mathbb{Z}/n$	Complete (n odd); Gap (n even)

Single most important gap to close first. Proposition 5.6/Remark 5.7 for *even* n : prove that $(n/2)[\alpha]$ is not the inflated Schur class, equivalently that the transgression cocycle $\gamma \in H^1(S_n; A_{n-1}) = \mathbb{Z}/n$ of the extension (3) is a generator. This does not affect the isomorphism type of H^2 (already proved), only the statement that the *canonical* class is a generator of the $\mathbb{Z}/n$ summand rather than a generator-plus-Schur.

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7 Resolution of the even- n generation gap

Earlier drafts flagged, as open at even n , whether the canonical class c generates the $\mathbb{Z}/n$ summand of $H^2(\mathrm{PN}(T); \mathrm{U}(1)) \cong \mathbb{Z}/n \oplus \mathbb{Z}_2$. We close it. The two constituent facts are machine-checked in `verification/paperA_evenN.py`.

Lemma 7.1 (The order alone is not enough at $n \equiv 2 \pmod{4}$). *In $\mathbb{Z}/n \oplus \mathbb{Z}_2$ an element of order n need not have a generating first coordinate: for $n \equiv 2 \pmod{4}$ the element $(2, 1) \in \mathbb{Z}/6 \oplus \mathbb{Z}_2$ has order 6 while $\langle 2 \rangle \subsetneq \mathbb{Z}/6$. Thus “order n ” does not by itself place c in the standard $\mathbb{Z}/n$ summand.*

Proposition 7.2 (Zero Schur component). *The $\mathbb{Z}_2$ (Schur) summand of $H^2(\mathrm{PN}(T); \mathrm{U}(1))$ is $H^2(S_n; \mathrm{U}(1))$ embedded by inflation along the split projection $\mathrm{PN}(T) = (T/\mathrm{U}(1)) \rtimes S_n \rightarrow S_n$, and its complementary projection is restriction to the S_n subgroup. The symmetric group lifts to honest permutation matrices $P_\sigma \in N(T)$ with $P_\sigma P_\tau = P_{\sigma\tau}$ exactly (no scalar phase), so the central extension $\mathrm{U}(1) \rightarrow N(T) \rightarrow \mathrm{PN}(T)$ splits over S_n and $\mathrm{res}_{S_n}(c) = 0$. Hence the $\mathbb{Z}_2$ component of c vanishes, for every n .*

Theorem 7.3 (Generation, all n). *c generates the $\mathbb{Z}/n$ summand of $H^2(\mathrm{PN}(T); \mathrm{U}(1))$ for every n , in particular $n = 4$ (Peres–Mermin).*

Proof. By Proposition 7.2 the Schur component of c vanishes, so under the splitting $c = (a, 0)$. By the order theorem c has order exactly n , whence a has order n , i.e. a generates $\mathbb{Z}/n$; therefore $\langle c \rangle = \mathbb{Z}/n \oplus 0$ is the standard summand. $\square$

Residual (one textbook step). The argument uses one standard, non-elementary fact: for a *split* extension the restriction to the complement S_n retracts the inflation of $H^2(S_n)$ — so res_{S_n} is exactly the $\mathbb{Z}_2$ -projection — and annihilates the $E_\infty^{1,1} = \mathbb{Z}/n$ part. This is the usual behaviour of the Lyndon–Hochschild–Serre edge maps for a split extension; a specialist should confirm it in the smooth/topological setting. Modulo this identification the gap is closed.